

Extrusion-modulated DnaA activity oscillations coordinate DNA replication with biomass growth

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eLife Assessment

This work provides **valuable** insights by introducing a post-translational extrusion mechanism that could reshape how we understand the coupling between DnaA activity and DNA-replication initiation. While **solid** evidence is presented for some of the key results, other claims rest on indirect proxies and could be improved.

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Abstract

Robust control of DNA replication is fundamental to bacterial proliferation. In *Escherichia coli*, replication initiation is thought to be regulated by oscillations in DnaA activity, driven by DnaA-chromosome interactions that differ among leading models. However, direct evidence linking these oscillations to replication initiation has been lacking, and existing models fail to explain the observed decoupling of replication initiation from *dnaA* expression. Here, we establish a direct link between DnaA activity and replication initiation by demonstrating robust oscillations in DnaA activity, which peak precisely at replication initiation across diverse growth conditions and genetic perturbations. Notably, these oscillations persist even when *dnaA* transcription remains constant, suggesting a regulatory mechanism that modulates DnaA activity independently of its expression. Additionally, we propose an extrusion model in which DNA-binding proteins sense biomass-DNA imbalance and extrude DnaA from the chromosome to trigger replication, overcoming limitations of existing models. Consistent with this model, perturbation of the nucleoid-associated protein H-NS modulates DnaA activity and replication timing, supporting its mechanistic validity.

Introduction

Bacterial proliferation requires precise coordination between DNA replication and biomass growth to maintain cellular homeostasis. While biomass accumulates exponentially, DNA synthesis proceeds in a stepwise manner, creating temporal imbalances that demand continuous correction throughout the cell cycle (**Fig. 1A-B**). Failure to synchronize these processes risks catastrophic outcomes, such as growth defects, incomplete replication, unequal genetic segregation, and cell death (Boye et al., 1996; Khodursky et al., 2015; Lu et al., 1994; Mäkelä et al., 2024; Zaritsky & Pritchard, 1971). Central to this synchronization in *Escherichia coli* is the initiator protein DnaA, a DNA-binding protein that binds specifically to an asymmetric 9-bp DnaA-box, whose consensus sequence is TTWTNCACA (Hansen & Atlung, 2018; Speck et al., 1999). Although DnaA binding to the replication origin (*oriC*) triggers replication initiation (Dong et al., 2023; Kaguni, 2011; Ohbayashi et al., 2020; Ozaki & Katayama, 2012; Sekimizu et al., 1987), how DnaA activity is dynamically regulated to align replication initiation with cellular growth, particularly when DNA content and biomass diverge, remains a fundamental unresolved question in cell cycle biology.

Two competing models have dominated the field. The titration model posits that DnaA activity is governed by the availability of free DnaA, which accumulates until it surpasses the buffering capacity of chromosomal DnaA-boxes (Christensen et al., 1999; Hansen et al., 1991). Here, replication initiation resets the system by increasing DNA content (and thus DnaA-boxes), titrating excess DnaA. In contrast, the switch model emphasizes the ATP/ADP-bound states of DnaA: DnaA-ATP, synthesized during biomass growth, triggers initiation, while hydrolysis to DnaA-ADP following replication resets the cycle (Berger & Wolde, 2022; Donachie & Blakely, 2003; Fu et al., 2023; Hansen & Atlung, 2018; Katayama et al., 2017). Crucially, the titration model depends on ongoing DnaA synthesis to reflect biomass accumulation, whereas the switch model relies on nucleotide-state transitions, independent of total DnaA levels. Despite these mechanistic differences, both models consider that replication initiation is regulated by oscillations in DnaA activity, driven by DnaA-chromosome interactions (**Fig. 1C**). However, direct evidence linking these oscillations to replication initiation remains lacking. Moreover, neither model fully explains recent experimental observations: replication initiation persists for multiple generations even when *dnaA* transcription is completely inhibited (Knöppel et al., 2023). This discrepancy suggests an uncharacterized mechanism that senses biomass-DNA imbalance and directly modulates DnaA activity.

Here, using a synthetic reporter coupled with single-cell mRNA fluorescence *in situ* hybridization (FISH), we demonstrate that DnaA activity peaks precisely at initiation across diverse growth conditions and genetic backgrounds, even when *dnaA* transcription is held constant. DnaA activity oscillations persist independently of transcriptional feedback, pointing to a post-translational regulatory mechanism. We propose an extrusion model in which nucleoid-associated proteins (NAPs) sense biomass-DNA imbalance and dynamically extrude DnaA from the chromosome, liberating it to activate *oriC*. This model not only explains the decoupling of replication timing from *dnaA* expression but also integrates disparate experimental findings. Perturbations of the NAP, H-NS for instance, modulate DnaA activity and replication timing, directly supporting the model's predictions. By unifying regulatory principles across transcription, protein activity, and chromosome dynamics, our findings redefine the paradigm for bacterial replication control.

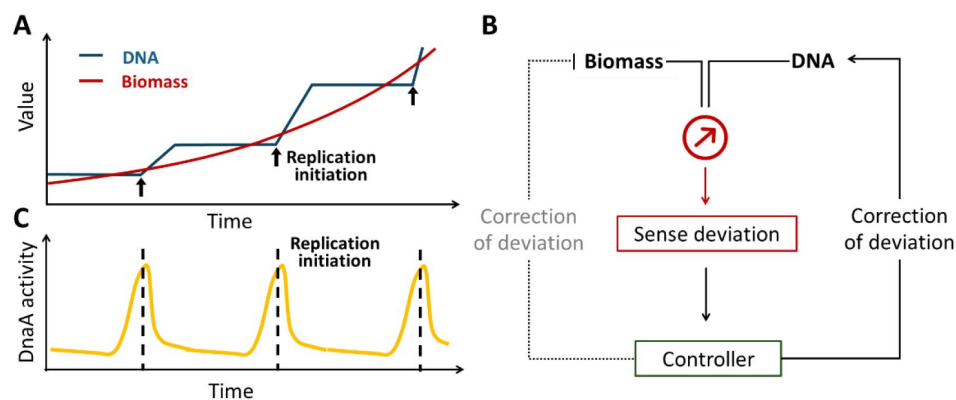


Figure 1.

Coordination of biomass growth and DNA replication through replication initiation and DnaA activity oscillations.

(A) Schematic representation of the discrepancy between biomass accumulation and DNA replication. While biomass grows exponentially, DNA synthesis progresses linearly, necessitating replication initiation events to maintain coordination. (B) Mechanistic model for biomass-DNA coordination in bacteria. A molecular sensor detects deviations between cell mass and DNA content, transmitting this information to regulatory controllers that compensate by either increasing DNA replication or restricting biomass accumulation. (C) Illustration of cyclic DnaA activity oscillations aligning with replication initiation to ensure precise cell cycle control.

Results

Development of a DnaA activity reporter system

It has been known that changing DnaA expression levels affects DnaA activity, and further changes initiation mass (Berger & Wolde, 2022 [DOI](#); Løbner-Olesen et al., 1989 [DOI](#); Zheng et al., 2020 [DOI](#)). To investigate this further, we engineered a *dnaA*-titratable strain, where *dnaA* transcription is controlled by an inducible P_{tet} promoter (Lutz & Bujard, 1997 [DOI](#)) (Fig. 2A [DOI](#)). This system enabled precise modulation of *dnaA* mRNA levels (0.25 to 6 times the wild-type level) via anhydrotetracycline (aTc) induction (0.75–20 ng ml⁻¹) without altering growth rate or cell size (Fig. 2B–D [DOI](#)). Strikingly, increasing DnaA expression doubled population-averaged *oriC* copy numbers and reduced initiation mass (the cell volume per *oriC* at replication initiation) by 50 % (Fig. 2E–F [DOI](#)). The observed relationship between DnaA expression and initiation mass deviated significantly from both the titration and switch models (Fig. 2G [DOI](#)), suggesting unaccounted regulatory mechanisms.

To dissect DnaA activity dynamics, we constructed a library of synthetic promoters ($n = 67$) by replacing *tetO* operators in P_{tet} with DnaA-boxes (Fig. S1). To assess the degree of DnaA activity-mediated repression, we measured the fold change in GFP expression driven by each synthetic promoter under low and high levels of *dnaA* induction (Fig. 3A [DOI](#)). Six synthetic promoters exhibited over 8-fold DnaA-mediated repression, whereas the promoter lacking DnaA-boxes (P_{con}) showed none, and the well-characterized native *dnaA* promoter (P_{native}) (Braun et al., 1985 [DOI](#); Saggioro et al., 2013 [DOI](#); Speck et al., 1999 [DOI](#)) displayed approximately two-fold repression (Fig. 3B [DOI](#); Fig. S1). P_{syn66} , which contains three strong and one weak DnaA-box, exhibited >8-fold repression under high DnaA induction (Fig. 3B [DOI](#)). P_{syn66} 's design enables responsiveness to both free DnaA levels (via strong boxes) and the ATP/ADP ratio (via differential binding to the weak box) (Grimwade & Leonard, 2021 [DOI](#); Katayama et al., 2017 [DOI](#); Speck et al., 1999 [DOI](#)). Dose-response assays confirmed that P_{syn66} was progressively repressed across a 40-fold range of *dnaA* expression (Fig. 3C [DOI](#)), with no interference from SeqA-mediated sequestration (Fig. 3D–E [DOI](#)). These results validated P_{syn66} as a robust and specific reporter of DnaA activity.

Decoupling DnaA activity oscillations from *dnaA* transcription fluctuations

To investigate DnaA activity dynamics throughout the cell cycle, we linked P_{syn66} activity to cell size, a surrogate for cell-cycle progression, using single-cell mRNA FISH (Skinner et al., 2013 [DOI](#)) to quantify *lacZ* mRNA levels in MG1655Δ*lac* cells harboring the P_{syn66} -*lacZ* plasmid (Fig. 4A [DOI](#)). A control strain harboring a promoter-less *lacZ* construct (P_{neg}) showed undetectable fluorescence signals (Fig. 4A [DOI](#)), confirming the specificity of *lacZ* mRNA detection. Quantitative analysis revealed that P_{syn66} -driven mRNA levels fluctuated approximately 3-fold over the cell cycle, whereas the DnaA-unresponsive constitutive promoter (P_{con}) exhibited stable expression (Fig. 4B [DOI](#)). However, fluctuations in mRNA concentration may not directly correspond to changes in DnaA activity due to global factors such as plasmid copy number and RNA polymerase concentration (Balakrishnan et al., 2022 [DOI](#); Bintu et al., 2005a [DOI](#); Bintu et al., 2005b [DOI](#)). To account for these variables, we used a P_{con} -*lacZ* plasmid to estimate global effects on mRNA concentration and calculated DnaA activity as inversely proportional to $k_{syn66} = \frac{[mZ](P_{syn66})}{[mZ](P_{con})}$. In wild-type cells (MG1655Δ*lac*) harboring the synthetic reporter system (Fig. 4C [DOI](#)), fluctuations in DnaA activity, as denoted by k_{syn66}^{-1} , displayed approximately a 3-fold variation across the cell cycle (Fig. 4D [DOI](#)), underscoring the substantial cell cycle-dependent oscillations in DnaA activity.

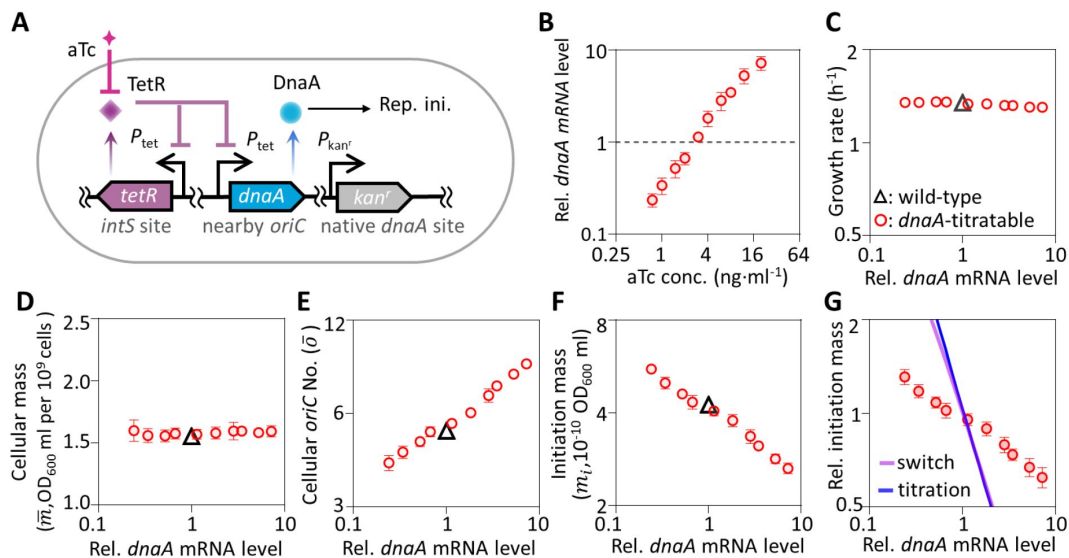


Figure 2.

Construction and characterization of the *dnaA*-titratable strain.

(A) Schematic of the *dnaA*-titratable strain. A *dnaA* gene under the control of the P_{tet} promoter was inserted near *oriC* and regulated by a P_{tet} -*tetR* feedback loop integrated at the *intS* locus, enabling fine-tuned expression control. The native *dnaA* gene was replaced with a kanamycin resistance cassette (*kan^r*). **(B–F)** Characterization of *dnaA*-titratable cells (red open circles) grown in rich defined medium with glycerol (M6) under varying aTc concentrations. Measured parameters include *dnaA* mRNA levels **(B)**, growth rate **(C)**, population-averaged cellular mass **(D)**, population-averaged *oriC* numbers **(E)**, and initiation mass **(F)**. Wild-type MG1655 values (black triangles) are shown for comparison. *dnaA* mRNA levels, normalized to wild-type MG1655, increased with aTc concentration from 0.75 to 20 ng·ml⁻¹. Cellular mass was determined by OD₆₀₀ divided by cell number concentration. *oriC* copy numbers were measured using a run-out assay, and initiation mass was calculated as the ratio of cellular mass to *oriC* numbers. Data represent means ± SD (n = 5 biological replicates). **(G)** Relationship between relative initiation mass and relative *dnaA* mRNA levels, compared with predictions from the initiation titration model (blue line) and the switch model (purple line). Experimental data are overlaid for validation.

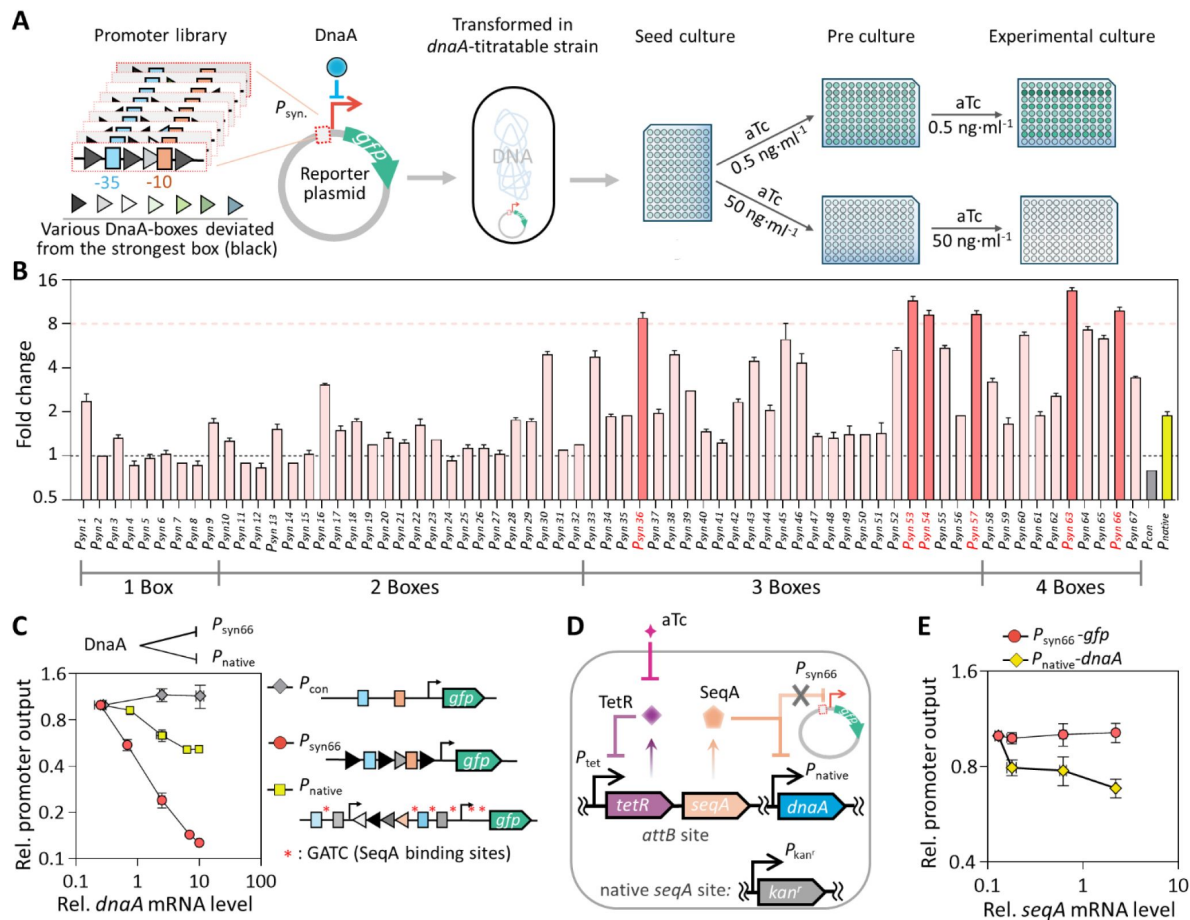


Figure 3.

Development of a DnaA activity reporter system.

(A) Schematic of promoter design and screening. Sixty-seven synthetic promoters were constructed by inserting various DnaA-boxes around the promoter core to drive *gfp* expression in a *dnaA*-titratable strain, where *dnaA* expression was regulated by aTc concentration. After pre-cultivation, GFP fluorescence per OD₆₀₀ was measured using a microplate reader in cells grown under low (0.5 ng·ml⁻¹) or high (50 ng·ml⁻¹) aTc concentrations to determine repression fold-change. **(B)** Repression fold-change of synthetic promoters. P_{con} (a promoter lacking DnaA-boxes) and P_{native} (the endogenous *dnaA* promoter) served as negative and positive controls, shown in gray and yellow, respectively. **(C)** Response curves of three promoters to varying *dnaA* expression levels, with their promoter architectures shown on the right. Promoter activity was assessed by relative *gfp* mRNA levels, normalized to the lowest *dnaA* expression condition. Schematic **(D)** and response curves **(E)** of P_{syn66} and P_{native} responses to SeqA in the *seqA*-titratable strain. Promoter activity was quantified from *gfp* transcript levels in *seqA*-titratable cells containing the P_{syn66}-GFP plasmid, normalized to the lowest *seqA* expression level. All cells were grown in rich defined medium supplemented with glycerol across different aTc concentrations. Data represent mean ± SD from 3 biological replicates **(B,C,E)**.

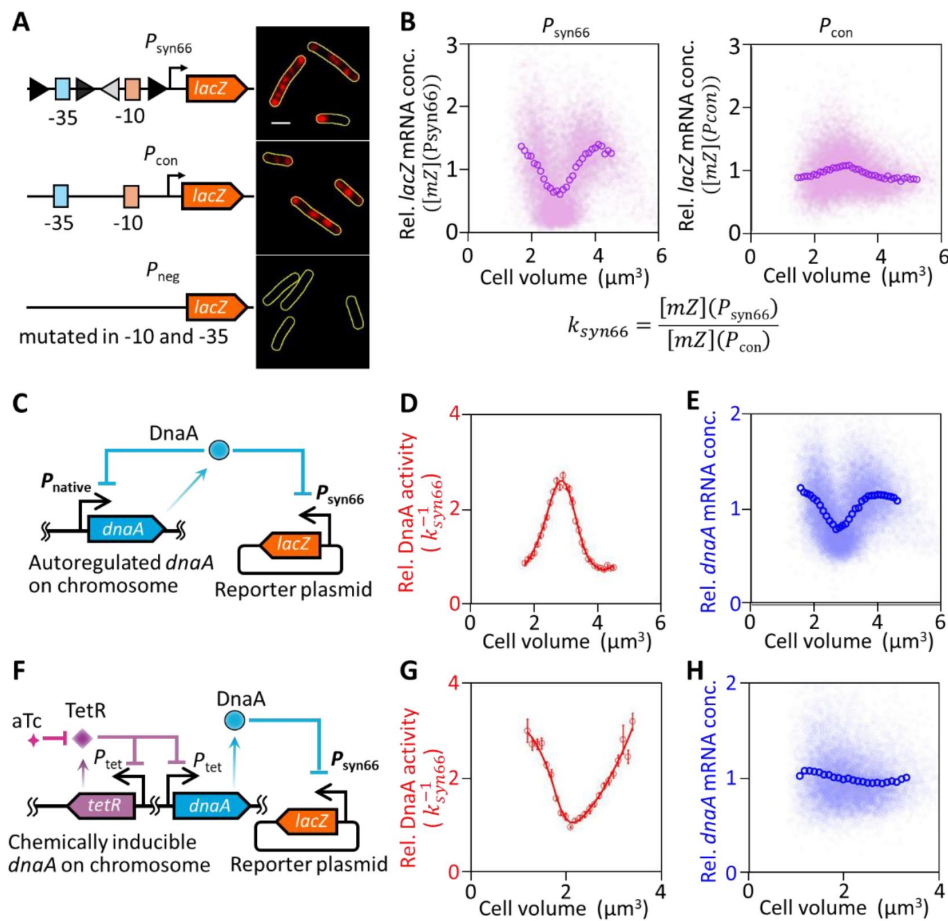


Figure 4.

DnaA activity oscillations decoupled from *dnaA* transcription fluctuations.

(A) Representative *lacZ* mRNA FISH images of MG1655 Δlac cells transformed with *lacZ* expression plasmids driven by P_{syn66} (DnaA-boxes around promoter core), P_{con} (no DnaA-box around promoter core), or P_{neg} (mutated promoter core). Yellow outlines indicate cell boundaries identified from phase-contrast images. **(B)** Relative *lacZ* mRNA concentrations driven by P_{syn66} (left) and P_{con} (right) across different cell volumes. Relative concentrations were determined from volume-specific *lacZ* mRNA fluorescence intensities, normalized to the population average. Volume-binned data for P_{syn66} ($[mZ](P_{syn66})$) and P_{con} ($[mZ](P_{con})$) are shown as open circles and were used to calculate k_{syn66} . **(C)** Schematic of a strain with autoregulated *dnaA* transcription carrying a DnaA activity reporter plasmid. Cell cycle-dependent fluctuations in relative DnaA activity **(D)** and relative *dnaA* mRNA concentrations **(E)** in cells from panel **C**, grown in rich defined medium supplemented with glucose. Relative DnaA activity (k_{syn66}^{-1}), calculated from volume-binned data in panel **B**, was smoothed and plotted as a red curve **(D)**. Relative *dnaA* mRNA concentrations were determined from volume-specific *dnaA* mRNA fluorescence intensities, normalized to the population average, with volume-binned data shown as open circles **(E)**. **(F-H)** Same as panels **C-E**, but for cells with aTc induced *dnaA* transcription, grown in rich defined medium supplemented with glycerol under 2 ng·ml⁻¹ aTc induction. More than 8,000 cells were analyzed per growth condition, and all error bars correspond to standard error of the mean (SEM).

We then examined whether the observed fluctuations in DnaA activity were linked to oscillations in *dnaA* transcription. Comparing the dynamics of *dnaA* mRNA and DnaA activity, we found that the peak in DnaA activity coincided with the trough in *dnaA* mRNA levels (**Fig. 4D-E**). To test whether *dnaA* transcription drives these oscillations, we eliminated transcriptional fluctuations using the *dnaA*-titratable strain. Despite constant *dnaA* mRNA levels (2 ng ml⁻¹ aTc induction), DnaA activity was retained robust oscillations (**Fig. 4F-H**), demonstrating the decoupling of DnaA activity oscillations from transcriptional fluctuations. These suggest that post-translational regulation, not transcriptional feedback, governs the DnaA activity.

DnaA activity oscillations are tightly coupled to replication timing

To explore the relationship between DnaA activity oscillations and DNA replication initiation, we examined cells grown under various nutrient conditions supporting different doubling times (30–66 min). Significant fluctuations in DnaA activity were observed across all conditions (**Fig. 5A**). We also estimated the cell volume at replication initiation (V_i) based on population-averaged cell volume and *oriC* number (Bremer et al., 1979; Si et al., 2017; Zheng et al., 2016) (Fig. S2 A–C), marking this on the DnaA activity profiles. In a representative birth-to-division cell cycle (Pountain et al., 2024) (Fig. S2D), DnaA activity peaks consistently coincided with V_i , indicating a close correlation between DnaA activity and replication initiation (**Fig. 5A**).

Further genetic perturbations, such as titrating DnaA expression, shifted DnaA activity fluctuations, yet the relationship between V^* (the cell volume at maximal DnaA activity in a representative birth-to-division cell cycle) and V_i remained locked (**Fig. 5B**). To quantify this relationship, we compared V_i with V^* , revealing a strictly proportional relationship through the origin (slope = 1.0, $R^2 = 0.98$), indicating equivalence between V^* and V_i (**Fig. 5C**). These results confirm that DnaA activity oscillations are tightly correlated with replication initiation, independent of transcriptional or nutrient perturbations.

An extrusion model reconciles paradoxical initiation events

Very recently, Knöppel *et al.* reported that multiple rounds of initiation occurred even after DnaA synthesis is shut down (Knöppel et al., 2023). In their experiment, DnaA was supplied in excess before expression ceased (Knöppel et al., 2023), potentially allowing continued replication initiation due to residual DnaA. To address this issue, we shut down DnaA expression from its native level using the deactivated miniature Un1Cas12f1 (dUn1Cas12f1) system, guided by sgRNA to repress *dnaA* transcription (**Fig. 6A**). This system effectively diminished *dnaA* transcription to an extremely low level, yet total *oriC* numbers increased fourfold within 90 min, consistent with two rounds of replication initiation (**Fig. 6B**). These findings demonstrate that halting DnaA synthesis does not immediately abolish replication initiation.

Both the titration and switch models predict a close relationship between DnaA activity oscillations and replication initiation (Berger & Wolde, 2022). However, they fail to account for the multiple rounds of initiation observed after DnaA synthesis was shut down. According to the titration model, initiation should cease immediately once DnaA expression is shut down, as the reduction in DnaA concentration would prevent it from reaching the threshold required to trigger new rounds of replication. In contrast, the switch model predicts infinite initiations following the cessation of DnaA expression (**Fig. 6C**). This is because the cessation of DnaA synthesis only reduce the production rate of DnaA-ATP and had no effect on the conversion rates between DnaA-ATP and DnaA-ADP. As a result, the ratio of DnaA-ATP to DnaA-ADP would temporarily decrease, causing a brief lag before initiation resumes indefinitely (**Fig. 6C**).

The failure of both models prompts us to revisit the regulatory mechanisms of replication initiation. In the titration model, a substantial fraction of DnaA proteins binds to DnaA-boxes along the chromosome, raising the possibility that these bound proteins serve as a reservoir of free DnaA, sustaining replication initiation even after DnaA synthesis ceases. We thus considered the

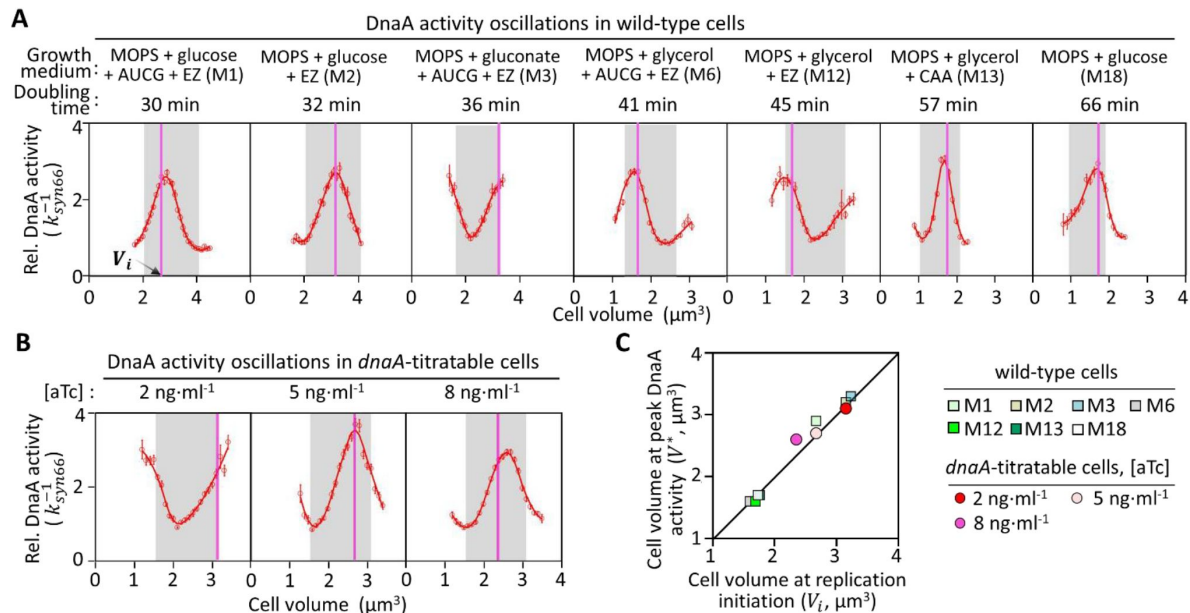


Figure 5.

Tight correlation between DnaA activity oscillations and DNA replication initiation.

(A) Cell cycle-dependent DnaA activity oscillations in wild-type cells across various growth conditions. DnaA activity is represented by k_{syn66}^{-1} . Volume-binned DnaA activity (red circles) was smoothed and plotted as a red curve. The representative birth-to-division cell cycle, defined as the cell volume doubling interval containing the majority of cells, is shaded in gray. The vertical line indicates the cell volume at replication initiation (V_i). **(B)** Cell cycle-dependent DnaA activity oscillations in *dnaA*-titratable cells grown in M6 medium with varying aTc concentrations. More than 8,000 cells were analyzed per growth condition, and error bars show the mean $\pm$ SEM. **(C)** Correlation between V^* and V_i in wild-type (squares) and *dnaA*-titratable (circles) cells. V^* represents the cell volume at the peak of DnaA activity within the representative birth-to-division cell cycle. The black line indicates equivalence between V^* and V_i .

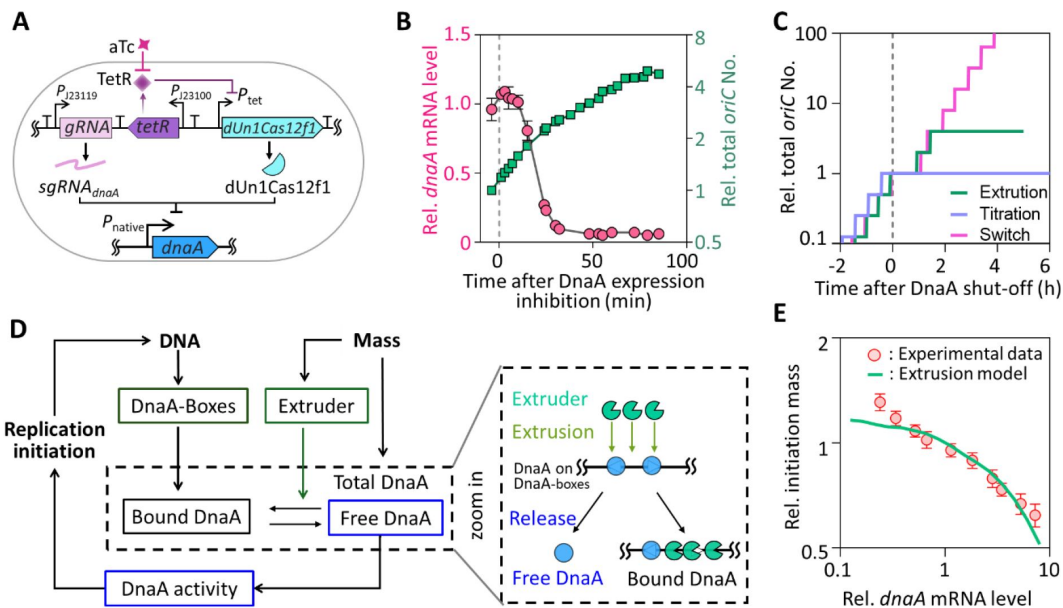


Figure 6.

An extrusion model explains DnaA shut-down dynamics.

(A) Genetic circuit of the deactivated CRISPR-Cas system for *dnaA* transcription shut-down. *dnaA* gene is targeted by a constitutively expressed sgRNA, while dUn1Cas12f1 expression is inhibited by TetR repressor. These transcription units are separated by terminators. The cassette was integrated into the chromosome near the *oriC* locus. DnaA shutdown is induced by the addition of aTc. (B) Time-course of relative *dnaA* mRNA levels (red line, left axis) and total *oriC* number (green line, right axis) following the addition of 50 ng·ml⁻¹ aTc at time 0 (dashed line). *dnaA* mRNA levels were normalized to wild-type levels, and *oriC* numbers were normalized to their initial values. Error bars indicate mean ± SD (n = 3 biologically independent experiments). (C) Predicted increases in total *oriC* number during *dnaA* transcription shut-down based on three models: the titration model, switch model, and extrusion model. Shutdown was simulated by setting *dnaA* transcription to zero at time 0 (dashed line). (D) Schematic of the extrusion model. The model introduces extruder(s) as additional regulators of biomass-DNA coordination, complementing the role of DnaA (left). Increased binding of the extruder to DNA promotes the release of DnaA from DnaA-boxes (right). (E) Comparison of the relationship between relative initiation mass and relative *dnaA* mRNA levels from experimental data (Fig. 2F) and predictions of the extrusion model.

possibility of some extruder(s) alongside DnaA to sense biomass-DNA deviation, and flush DnaA proteins away from the DnaA-boxes, thus releasing free DnaA to dynamically restore DnaA activity, even in the absence of ongoing DnaA synthesis (Fig. 6D [↗](#)).

In this proposed “extrusion model”, the extruder is constitutively expressed, synthesized at a constant rate throughout the cell cycle, accumulating in proportion to biomass growth, enabling replication initiation when free DnaA concentrations exceed the critical threshold, as described in the titration model (Hansen & Atlung, 2018 [↗](#); Hansen et al., 1991 [↗](#)) (Fig. 6D [↗](#)). This model guarantees the long-term coordination of biomass growth and *oriC* abundance, as well as stable oscillations in free DnaA and *oriC* concentrations (Fig. S3 A). Notably, the extrusion model accounts for the persistence of replication initiation and free DnaA oscillations over multiple cell cycles after DnaA synthesis is halted (Fig. 6C [↗](#); Fig. S3 B). Moreover, since the extruder mitigates the impact of perturbations on DnaA synthesis, the model predicts a quantitative relationship between initiation mass and DnaA expression levels that aligns with experimental observations (Fig. 6E [↗](#)). Furthermore, a stochastic implementation of the extrusion model shows the lack of correlation between successive replication initiation events (Fig. S3 C), as observed in recent studies (Si et al., 2019 [↗](#); Witz et al., 2019 [↗](#)).

Perturbation of H-NS validates the extrusion mechanism

The extrusion model could be validated by changing the expression level of the extruder, which modulates DnaA activity by releasing DnaA from DnaA-boxes. Given their DNA binding properties, NAPs (Dillon & Dorman, 2010 [↗](#)) are likely candidates for this extruder role. Therefore, we selected H-NS (Dorman, 2004 [↗](#)), a major NAP, which is capable of promoting the release of DnaA from bound DnaA-boxes *in vitro* (Fig. S4), to experimentally assess the validity of the extrusion model. To this end, we constructed a *hns*-titratable strain, incorporating a reporting plasmid to capture changes in DnaA activity across various H-NS expression levels (Fig. 7A [↗](#)). Under steady-state cultivation, increasing aTc concentrations from 0 to 50 ng·ml⁻¹ resulted in *hns* mRNA levels ranging from 0.25 to 5 times the wild-type levels, without affecting growth rate (Fig. S5 A-B). Within this range, DnaA activity increased by approximately 60 % (Fig. 7B [↗](#)), while the initiation mass decreased by 30 % (Fig. 7C [↗](#); calculated with Fig. 5S C-D [↗](#)), in qualitative agreement with the predictions of the extrusion model (Fig. S5 E-F).

Since H-NS is a global transcriptional regulator (Hommais et al., 2001 [↗](#)), we wonder whether its effect on DnaA activity was mediated through *dnaA* transcription. To test this, we measured DnaA activity and *dnaA* mRNA levels during an H-NS shift-up. Upon addition of 50 ng·ml⁻¹ aTc, *hns* levels increased more than 20-fold within a short period, followed by approximately 80 % increase in DnaA activity, with no significant change in *dnaA* mRNA levels (Fig. 7D [↗](#)). Additionally, cellular *oriC* content increased significantly within 30 minutes, while cell mass remained unchanged (Fig. 7E [↗](#)). These results suggest that the rapid increase in DnaA activity and accelerated replication initiation induced by H-NS upregulation is not driven by changes in *dnaA* transcription.

Leveraging the *hns*-titratable strain, we re-examined the tight correlation between DnaA activity oscillations and replication initiation. In two different growth media with varying *hns* levels, the peak of DnaA activity fluctuations (V^*) shifted in concert with the independently measured volume at replication initiation (V_i) (Fig. 7F [↗](#)). Plotting V^* against V_i revealed the same equivalency observed in wild-type and *dnaA*-titratable cells (Fig. 7G [↗](#)). Collectively, these results support the notion that DnaA activity oscillations, modulated by extrusion, coordinate DNA replication and biomass growth.

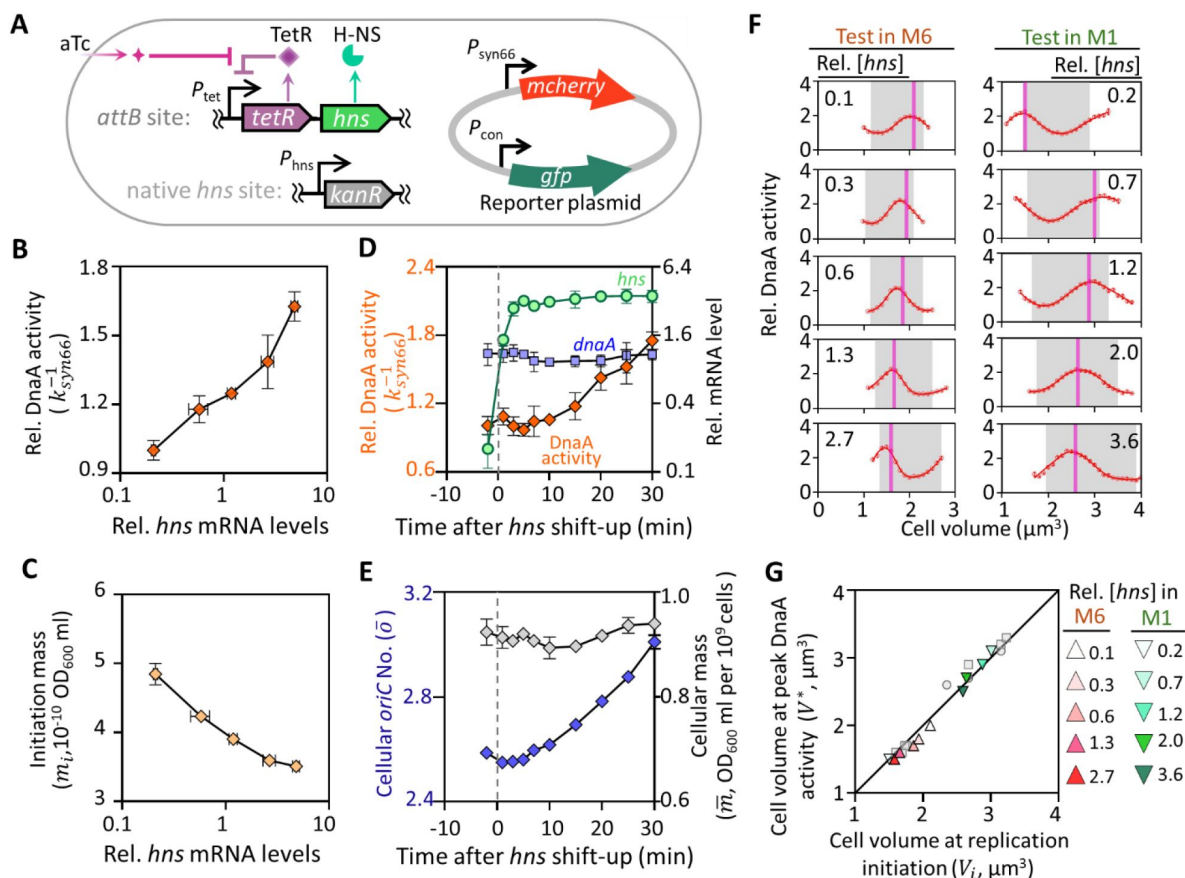


Figure 7.

Titration of *hns* expression modulates DnaA activity and replication initiation.

(A) Genetic circuit of the *hns*-titratable strain. Expression of *hns* is controlled by a P_{tet} -*tetR* negative feedback loop integrated at the *attB* site, with the native *hns* coding sequence replaced by a kanamycin resistance gene. The plasmid contains P_{syn66} -*mcherry* and P_{con} -*gfp* expression cassettes was used to assess DnaA activity. DnaA activity **(B)** and initiation mass **(C)** were characterized in M6 medium with varying *hns* expression levels during steady-state cultivation. **(D)** mRNA levels of *hns* (green circles) and *dnaA* (blue squares) relative to wild-type levels, along with DnaA activity (orange rhombus), were measured during *hns* shift-up. *hns* shift-up was induced by the addition of 50 ng·ml⁻¹ aTc at time 0 (dashed line), following steady-state cultivation in M6 medium without aTc. **(E)** Dynamics of population-averaged *oriC* number (blue rhombus, left axis) and cellular mass (gray rhombus, right axis) during *hns* shift-up. Data represent mean $\pm$ SD from 3 biological replicates **(B-E)**. **(F)** Cell cycle-dependent DnaA activity oscillations in *hns*-titratable cells cultivated in M1 and M6 media under varying aTc concentrations. Relative *hns* mRNA levels are indicated for each condition. More than 8,000 cells were analyzed for each condition and error bars show the mean $\pm$ SEM. **(G)** Correlation between the volume at maximal DnaA activity (V^*) and the volume at replication initiation (V_i) for *hns*-titratable cells grown in M1 (green down-pointing triangle) and M6 (red up-pointing triangle) media with varying *hns* expression levels. Data for *dnaA*-titratable cells (gray circles) and wild-type *dnaA*-autoregulated cells (**Fig. 5C**) are included for comparison.

Discussion

In this study, we proposed chromosome-driven DnaA activity oscillations as a post-translational “rheostat” coordinating DNA replication with biomass growth in *E. coli*. By developing a SeqA-independent synthetic promoter reporter (P_{syn66}) on a plasmid, we circumvented confounding factors inherent to native systems (Pountain et al., 2024), thereby achieving the direct quantification of DnaA activity dynamics. Crucially, these oscillations persist when *dnaA* transcription is fixed, operate independently of SeqA-mediated repression, and peak precisely at replication initiation across all tested growth conditions. This reveals an essential paradigm: bacteria bypass transcriptional control to synchronize genome duplication with growth through spatial regulation of DnaA-chromosome interactions.

Very recently, Iuliani *et al.* also reported DnaA activity oscillations (Iuliani et al., 2024), which is a significant milestone, innovatively using a chromosomal promoter-reporter system and microfluidics for single-cell DnaA activity observations. They revealed key relationships between DnaA activity, cell size, and division. Our work, however, differs in multiple ways. Methodologically, we engineered a *dnaA*-titratable strain and synthetic promoters like P_{syn66} , enabling precise *dnaA* expression control and cleaner DnaA activity reporting. Our mRNA FISH approach uniquely compares DnaA activity and transcription at the single-cell level. By decoupling initiation volume measurement from DnaA monitoring, we resolve a 1:1 coupling between DnaA activity thresholds and replication initiation events. Regarding research questions, while Iuliani *et al.* focused on DnaA-cell size-division relationships, we addressed the decoupling of replication initiation from *dnaA* expression with our novel extrusion model.

Recently, an integrated “titration-switch” model has been proposed to improve replication cycle stability across different growth rates (Berger & Wolde, 2022; Fu et al., 2023). Although this integrated model predicts a relationship between initiation mass and *dnaA* mRNA levels that aligns closely with steady-state experimental data (Fig. S6 A), it fails to account for the additional replication initiation observed after *dnaA* shutdown (Fig. S6 B). By incorporating an extruder into this model, we successfully reproduced the additional replication initiation (Fig. S6 C), thus validating the extrusion concept and aligning with experimental observations. These findings highlight the critical role of extrusion in refining our mechanistic understanding of biomass-DNA coordination.

Mechanistically, we propose that NAPs sense biomass-DNA imbalances and extrude DnaA from chromosomal sites, converting physical displacement into biochemical activation, a spatial control strategy mirroring eukaryotic chromatin regulation. H-NS emerges as a prime extruder candidate given its DNA-condensing ability (Dame et al., 2000; Zimmerman, 2006) and replication-timing effects (Fig. 7). This functional multiplicity likely enhances robustness, allowing cells to maintain replication homeostasis despite fluctuating environments, an evolutionary advantage paralleling eukaryotic checkpoint system.

These findings redefine bacterial cell cycle control as a dynamic interplay between physical chromosome remodeling and metabolic sensing. By tethering DnaA activation to biomass accumulation through extrusion, cells achieve a critical feat: noise suppression via spatial filtering of transcriptional fluctuations. Such mechanistic parsimony, using chromosome structure to both store genetic information and regulate replication timing, may represent a fundamental principle of cellular organization. Indeed, parallels to eukaryotic systems are striking: DnaA extrusion resembles ORC licensing through chromatin accessibility (Fragkos et al., 2015), while biomass-DNA coupling evokes yeast size-control checkpoints (Xie et al., 2022).

Future studies could map the complete extruder interactome and elucidate how metabolic signals modulate their activity. Our synthetic reporter system provides a blueprint for dissecting the spatial regulation of DNA-binding proteins, with translational potential in antimicrobial development (targeting pathogen-specific extruders) and synthetic biology (engineering tunable replication circuits). More broadly, this work establishes physical chromosome remodeling as a central coordinator of cellular homeostasis, a concept bridging prokaryotic and eukaryotic cell cycle paradigms.

Materials and Methods

Plasmid and strain Construction

Strains used in this study were derived from *E. coli* MG1655. Information for all strains and plasmids are listed in Tables S1 and S2.

Plasmids for reporting DnaA activity

Plasmids expressing GFP, mCherry, or LacZ under synthetic promoters were constructed based on the pPT plasmid backbone (Zong et al., 2017). The *lacZ* amplified from the MG1655 genome and *mcherry* gene amplified from pMD19-*hupA-mcherry* plasmid, were assembled into the BspQI/BsrGI-digested pPT backbone via the One Step Cloning Kit (Vazyme, C112), to generate the pPT-*lacZ* and pPT-RFP plasmids, respectively. Synthetic promoters, generated by annealing the primer pairs listed in Table S3, were ligated directly into the *BsaI*-digested pPT, pPT-RFP or pPT-*lacZ* plasmid as required. The native *dnaA* promoter (P_{native}) amplified from the MG1655 genome was digested with *BsaI*, and ligated into the *BsaI*-digested pPT plasmid. The $P_{\text{con}}\text{-gfp}$ cassette amplified from $P_{\text{con}}\text{-GFP}$ plasmid was assembled into the *SpeI*-digested $P_{\text{syn66}}\text{-RFP}$ plasmid to generate the $P_{\text{syn66}}\text{-}P_{\text{con}}\text{-FPs}$ plasmid. Primer information for amplifying DNA fragments is detailed in Table S4.

Plasmids for strain construction

For constructing the plasmid pMD19-Rhns, pMD19-RseqA and pMD19-RdnaA, the *hns*, *seqA* and *dnaA* gene were amplified from MG1655 genome, digested, and ligated into the same enzyme-digested pMD19-tetR plasmid. For constructing the plasmid p15A-RdnaA, the T3- $P_{\text{tet}}\text{-tetR-dnaA}$ cassette, *kan^r* gene and p15A-T1 fragment were amplified from plasmid pMD19-RdnaA, pEcCas and PZA31- $P_{\text{tet}}\text{-M2-GFP}$, respectively, and then assembled via MultiS One Step Cloning Kit (Vazyme, C113). The plasmid P_CRidnaA1 were obtained by ligating the annealed primer pairs to the *BsaI*-digested CPP00458 plasmid for the *dnaA* targeting. The chloramphenicol resistance gene *Cm^r* amplified from the PZA31- $P_{\text{tet}}\text{-M2-GFP}$ plasmid replaced the *kan^r* gene in the pEcCas plasmid via λ Red recombination method, resulting in the *CmPcas* plasmid. Primer information is detailed in Table S4.

Strain construction

The *seqA*-, and *hns*-titratable strains were constructed as previously described (Zheng et al., 2016). Briefly, after transforming pSIM5 plasmid into MG1655 cells, the $P_{\text{tet}}\text{-tetR-driven seqA}$ or *hns* cassette was inserted at the *attB* site on the genome, and then the native gene was replaced with the *kan^r* gene amplified from plasmid p15A-T1.

For the construction of *dnaA*-titratable strain, we first inserted the $P_{\text{tet}}\text{-tetR cassette}$ into the *intS* site via CRISPR/Cas9- λ Red genome editing system (Jiang et al., 2015) by targeting 20-nucleotide sequence (ttctcctgcagaccagatcc). Then applying the λ Red recombination systems in the *CmPcas*

plasmid, the *bla*: P_{tet} -*dnaA* cassette amplified from the pMD19-RdnaA plasmid via overlap PCR, was inserted between the *gidA* and *gidX* genes, and the negative P_{native} -*dnaA* was replaced with the T_3 -Kan- T_1 cassette amplified from the p15A-RdnaA plasmid.

In order to delete the native *lacZ* gene in the genome of the RdnaA1, Rhns1 and MG1655 cells, the homologous recombination fragment amplified from the genome of CL1 strain, was used to knock out the native *lac* operon via CRISPR/Cas9- λ Red editing system (Jiang et al., 2015) by targeting 20-nucleotide sequence (cttccggctcgatgtgttg).

For the construction of DnaA shut-down strain (CRidnaA1), the P_{J23119} -sgRNA_{dnaA}: P_{J23100} -*tetR*: P_{tet} -*dUn1Cas12f1* cassette was amplified from the P_CRidnaA1 plasmid and flanked with homologous sequences to the *asnA* and *viaA* gene. This cassette was then inserted into the MG1655 genome using

CRISPR/Cas9- λ Red editing system, targeting the 19-nucleotide sequence (aagccgcctgctcagacgc). The CPP00458 plasmid, kindly provided by the Xiongfei Fu lab, contains the Un1Cas12f1 variant, which carries mutations D143R, T147R, D326A, and K330R that abolish its DNA-cleaving activity. However, it retains the ability to bind DNA in a sgRNA-guided manner, thereby inhibiting the expression of the target gene.

All primers for the construction of DNA fragments are detailed in Table S4.

Growth conditions

Cells for strain construction were grown in Luria-Bertani (LB) medium with appropriate antibiotics: ampicillin (100 $\mu\text{g}\cdot\text{ml}^{-1}$), kanamycin (50 $\mu\text{g}\cdot\text{ml}^{-1}$), spectinomycin (50 $\mu\text{g}\cdot\text{ml}^{-1}$), chloramphenicol (25 $\mu\text{g}\cdot\text{ml}^{-1}$), based on plasmid presents. For titratable strains, 10 $\text{ng}\cdot\text{ml}^{-1}$ anhydrotetracycline (aTc) was added during strain construction and seed culture preparation. Seed cultures were initiated by inoculating 3–6 individual colonies from an LB agar plate into 14 ml round-bottom test tubes containing 2 ml of the desired medium for overnight growth.

For physiological phenotypic characterization, all cells were cultivated at 37°C with shaking at 150 rpm in a water bath shaker. The *dnaA*-, *seqA*- and *hns*-titratable cells were characterized under varying aTc concentrations in rich defined medium supplemented with glycerol (M6). *hns*-titratable cells were also characterized in rich defined medium supplemented with glucose (M1). When strains contained reporter plasmid, 25 $\mu\text{g}\cdot\text{ml}^{-1}$ chloramphenicol was added. The autoregulated-*dnaA* strain (MGCL1) with a reporter plasmid, was tested in seven different media: M1 (MOPS + EZ + AUCG + Glucose), M2 (MOPS + EZ + Glucose), M3 (MOPS + EZ + AUCG + Gluconate), M6 (MOPS + EZ + AUCG + Glycerol),

M12 (MOPS + EZ + Glycerol), M13 (MOPS + CAA + Glycerol), and M18 (MOPS + Glucose). Detailed medium compositions were provided previously (Zheng et al., 2020). Steady-state growth was established as previously described (Zheng et al., 2020) by culturing cells for ~10 generations in exponential growth phase with ~15-fold dilution at OD₆₀₀ ~0.2, using the same medium supplemented with antibiotic or aTc as required. OD₆₀₀ measurements were taken at consecutive time points using a spectrometer (Genesys 10s, Thermo Fisher Scientific), and the growth rate (or doubling time) was calculated by fitting the data to an exponential growth curve. Samples for determination of population-averaged cell mass ($\bar{m}$) and cellular origin number ($\bar{o}$), quantification of mRNA levels, and mRNA fluorescence in situ hybridization (FISH) were collected when OD₆₀₀ reached ~0.2.

Measurements of population-averaged cellular mass

$(\bar{m})$, cellular *oriC* number ($\bar{o}$), and initiation mass (m_i)

The population-averaged cell mass ($\bar{m}$), cellular *oriC* number ($\bar{o}$), and initiation mass (m_i) were calculated as described previously (Zheng et al., 2020 [DOI](#)). Briefly, $\bar{m}$ was determined by dividing the OD₆₀₀ by the cell concentration (OD₆₀₀ ml per 10⁹ cells). For cell counting, samples were collected in precooled cell count buffer (0.9% NaCl with 0.12% formaldehyde, filtered with a 0.22 µm filter) and stained with 1 µg·ml⁻¹ DAPI in the same buffer. Cell number concentration was measured by counting the DAPI-stained particles using flow cytometry equipped with 405 nm laser (CytoFLEX, Beckman Coulter Life Sciences). The $\bar{o}$ was calculated based on the classical run-out experiments (Zheng et al., 2020 [DOI](#)). In brief, cell suspensions were treated with 500 µg·ml⁻¹ rifampicin and 30 µg·ml⁻¹ cephalixin to inhibit new initiation and cell division, respectively, and maintained for 2-3 mass doublings to complete ongoing replication. Cells were fixed with 70% precooled absolute ethanol, and treated with stain buffer (20mM Tris-HCl, pH 8.0, 130 mM NaCl, 10 ng·ml⁻¹ DAPI). Using the same flow cytometry, we measured the cellular origin distribution based on DAPI signaling, from which $\bar{o}$ was calculated. m_i was calculated based on the equation: $m_i = \frac{\bar{m}}{\bar{o}} \times \frac{1}{\ln 2}$ (Bremer et al., 1979 [DOI](#); Si et al., 2017 [DOI](#); Zheng et al., 2016 [DOI](#)).

Screening of the synthetic promoter library

After transforming the reporter plasmid into *dnaA*-titratable cells, the seed culture was cultivated in M6 medium with 10 ng·ml⁻¹ aTc, then diluted 1:180 in fresh medium containing either 0.5 or 50 ng·ml⁻¹ aTc and cultivated at 37 °C and 800 r.p.m. in a digital thermo-shaker (AOSHENG) using flat-bottom 96-well plates (Corning). Following approximately 4 hours of pre-cultivation, the cultures were diluted 1:180 in fresh medium with the same aTc concentration. The OD₆₀₀ and GFP fluorescence intensity were continuously monitored during cell growth using a Synergy H1 microplate reader (BioTek). The non-fluorescent strain (MG1655) cultured in 180 µl of M6 medium for each experiment was used to subtract the background fluorescence. For OD₆₀₀ between 0.1 and 0.2, the repression fold change was calculated by dividing the mean fluorescence intensity per OD₆₀₀ of the cells grown with 0.5 ng·ml⁻¹ aTc induction by the corresponding intensity for cells grown with 50 ng·ml⁻¹ aTc induction.

Quantitative real-time PCR

Quantitative real-time PCR was performed using a protocol similar to previously described (Zheng et al., 2016 [DOI](#)). Briefly, RNA was stabilized using Bacteria Protect Reagent (Qiagen) and extracted using RNeasy mini kit (Qiagen). 500 ng RNA was reverse transcribed (TAKARA), and the resulting cDNA samples were diluted 1:25 with PCR-grade water. Quantitative real-time PCR was carried out on a qTower3 (Analytikjena) or CFX Connect system (Bio-Rad), using the following program: 30 s at 95 °C, followed by 40 cycles of denaturation (5 s at 95 °C), annealing, and elongation (30 s at 58 °C). Primer sequences were listed in Table S4. The *rpoA* gene was used as the reference gene.

lacZ and *dnaA* mRNA fluorescence in situ hybridization (FISH)

FISH for *lacZ* or *dnaA* mRNA was performed using a method similar to previously described (Skinner et al., 2013 [DOI](#)). Briefly, antisense DNA oligo probes were designed against *lacZ* or *dnaA*, and synthesized with TAMRA labeling at the 3' end. The sequences of probes targeting *lacZ* or *dnaA* are either as previously described (Skinner et al., 2013 [DOI](#)) or listed in Table S5. 27 ml steady-state growth culture (OD₆₀₀ ~ 0.2) was fixed by direct adding 3 ml of 37 % (vol/vol) formaldehyde, following by incubation for 30 min at room temperature with gentle mixing using a nutator. The fixed cells were collected by centrifugation at 600 g for 8 minutes, and washed twice with 1 ml RNase-free PBS. The cell pellet was resuspended in 300 µl DEPC water, and 700 µl precooled absolute ethanol was added to permeabilize the cells for 1 hour at room temperature in

the nutator. After permeabilization, the cells were divided into two equal parts for characterization of *lacZ* or *dnaA* mRNA as needed. Hybridization was performed as described previously (Skinner et al., 2013 [DOI](#)), with the stringency of the hybridization and wash solutions adjusted by changing the concentration of formamide, 30 % for *dnaA* probes and 40 % for *lacZ* probes. After hybridization, the cells were washed with the appropriate wash solution and prepared for imaging.

Cell imaging and cell size parameters measurement

For imaging, 3 μ l of cell suspension was pipetted onto the center of a 24 $\times$ 60-mm coverslip. A thin 1.5 % agarose pad (prepared with 1xPBS) was then placed on top of the cell suspension droplet. Fluorescent and phase images were acquired using an inverted microscope (IX-83, Olympus) equipped with a 100x oil objective (Olympus), an automated xy-stage (ASI, MS2000), and a sCMOS camera (Prime BSI, photometrics). Fluorescence excitation was performed as previously described (Xia et al., 2022 [DOI](#)). Solid-state lasers (Coherent OBIS: 405–100 LX, 561–100 LS) and Semrock emission bandpass filters (445/40 nm for DAPI and 598/25 nm for RFP) were used for excitation and collecting fluorescent signals.

Images were aligned with phase contrast channel by selecting at least six pairs of common control points and applying a geometric transformation using the MATLAB `imwarp` function. This allowed for adjustment of the cell masks in the RFP or DAPI channel. A customized ImageJ -based image-processing package, MicrobeJ (v5.13l₁) plugin for Fiji (Ducret et al., 2016 [DOI](#)), was used to contour cells and calculate cell size parameters and fluorescence intensity, including the mean cell width, cell length, and cell area from phase-contrast images. The cell volume was calculated using a cylinder-plus-two-hemispheres geometry. The total RFP intensity, corrected for background subtraction per cell volume, was then used to estimate the concentration of the related mRNA level.

Characterization of cell cycle-dependent fluctuations in DnaA activity

Data filtering was first performed based on the intensity of the DAPI fluorescence signal. To eliminate non-cell regions resulting from incorrect segmentation of the phase contrast images, only DAPI-positive cells were included in the analysis. To further minimize the potential impact of variability in fixation, permeabilization, and wash efficiencies, which could affect the TAMRA fluorescence signal (representing the mRNA FISH signal), only cells with mean DAPI signals within the 95% confidence intervals were included in the subsequent analysis.

The concentration of the *lacZ* mRNA ($[mZ]$) can be expressed as $[mZ] = J/\delta_Z$, where δ_Z is the degradation rate and J is the mRNA synthesis flux. J is in turn given by the promoter on-rate (k), the RNAP concentration ($[RNAP]$), and the concentration of the *lacZ* gene ($[G_Z]$). Assuming that the factors δ_Z , $[RNAP]$, and $[G_Z]$ are the same between the two strains with P_{syn66} and P_{con} driving *lacZ*, then the ratio of the mRNA concentration obtained at each point of the cell cycle gives the ratio of the promoter on-rates, $k_{P_{syn66}}$ and $k_{P_{con}}$. Since a constitutive promoter is non-responsive to DnaA activity, k_{con} is independent of cell cycle and we can obtain DnaA activity regulated promoter activity as $k_{syn66} = \frac{[mZ](P_{syn66})}{[mZ](P_{con})}$, then the DnaA activity is denoted as k_{syn66}^{-1} .

The concentration of the *lacZ* mRNA were binned based on cell volume, with a bin size of 0.1 μ m³, was used to determine $[mZ](P_{syn66})$ and $[mZ](P_{con})$. The relation between $[mZ](P_{con})$ and cell volume was smoothed using the MATLAB `smooth` function, and from this, k_{syn66} was deduced. Based on the DnaA activity oscillations, the cell volume where DnaA activity reaches maximum (V^*) was calculated during a representative birth-to-division cell cycle. The cell volume

distribution was used to calculate the number of cells in each volume-doubling range. The range containing the highest cells was considered the representative birth-to-division cell cycle and was shaded in gray (Fig. S2 D). From this, V^* was obtained.

Calculation of cell volume at the time of replication initiation (V_i)

Cell volume at the time of replication initiation (V_{init}) was calculated based on population-averaged cell volume ($\bar{v}$), and the population-averaged cellular *oriC* number ($\bar{o}$). The equation used for this calculation is as follows:

$$V_i = v_i \times N_{oriC}^{init} = \frac{\bar{v}}{\bar{o}} \times \frac{1}{\ln 2} \times 2^{\lfloor \log_2 \bar{o} \rfloor}$$

where v_i is the initiation volume by definition, i.e. cell volume per *oriC* at the time of replication initiation (Bremer et al., 1979 [\[1\]](#); Si et al., 2017 [\[2\]](#); Zheng et al., 2016 [\[3\]](#)). $\bar{v}$ was obtained by analyzing the phase contrast images, $\bar{o}$ was obtained by run-out experiments (Zheng et al., 2020 [\[4\]](#)). N_{oriC}^{init} is the cellular *oriC* number at the time of replication initiation. $\lfloor \log_2 \bar{o} \rfloor$ denotes the floor function of $\log_2 \bar{o}$.

DnaA shut-down experiment

To shut-down DnaA expression, steady-state growth of the CRidnaA1 cells was first established after ~10 generations of exponential growth in M1 medium. At OD₆₀₀ between 0.15 and 0.2, 24 mL of cell culture was transferred to 12 mL of pre-warmed M1 medium containing 150 ng·mL⁻¹ aTc to induce dUn1Cas12f1 expression at time 0. Under the guidance of *sgRNA_{dnaA}*, the transcription of *dnaA* gene was inhibited. Samples were collected at various time points to determine population-averaged cellular mass ($\bar{m}$), cellular *oriC* number ($\bar{o}$), and to quantify mRNA levels. When the OD₆₀₀ reached around 0.2, the cultures were diluted about 2-fold with pre-warmed M1 medium containing 50 ng·mL⁻¹ aTc to maintain adequate nutrition. The steady-state growing MG1655 cells in M1 medium were also collected for quantification of wild-type *dnaA* mRNA levels. Three independent experiments were conducted.

H-NS shift-up experiment

The DnaA activity reporter plasmid P_{syn66}-P_{con}-FPs was transformed into the *hns*-titratable strain (Rhns2), and cells were cultivated overnight in M6 medium containing 10 ng·mL⁻¹ aTc and 25 µg·mL⁻¹ chloramphenicol for seed culture. Seed cultures were washed and grown in the same medium without aTc for ~10 generations of exponential growth. At an OD₆₀₀ of ~0.2, 40 mL of cell culture was transferred to 20 mL of pre-warmed same medium containing 150 ng·mL⁻¹ aTc to induce H-NS expression at time 0. Samples were collected at time points -2, 1, 3, 5, 7, 10, 15, 20, 25, and 30 minutes for the determination of population-averaged cellular mass ($\bar{m}$), cellular *oriC* number ($\bar{o}$), and quantification of mRNA levels. Steady-state growing MG1655 in M6 medium were also collected for *hns* mRNA quantification. Three independent experiments were conducted.

Electrophoretic mobility shift assay (EMSA)

The genes encoding *dnaA* and *hns* from the MG1655 genome were individually amplified and cloned into the expression vector pET-28a using primers listed in Table S4. *E. coli* BL21 (DE3) carrying pET-28a-DnaA or pET-28a-H-NS was used for overproduction of His₆-DnaA or His₆-H-NS. For each protein purification, a 1 L culture was induced by the addition of 1 mM IPTG at OD₆₀₀ = 0.4. After 1.5 hours of incubation, cells were harvested by centrifugation (5,000 rpm for 20 min at 4 °C). The pellet was resuspended in 50 ml of buffer A (50 mM Tris-HCl, pH 7.0, 500 mM NaCl, 0.5 mM PMSF, 5 % glycerol). Cell resuspension was lysed on ice for 30 minutes (3 seconds on, 7 seconds off) using a sonicator at 4 °C, and cell debris was removed by centrifugation (14,000 rpm for 60 min at 4 °C). Subsequently, the protein was purified from the soluble supernatant by Ni²⁺-affinity chromatography (HisTrap™ HP Columns; GE Healthcare) with a ÄKTA Pure Protein Purification

System. Proteins were collected using gradient elution with buffer B [20 mM Tris-HCl, pH 7.5, 500 mM NaCl, 400 mM imidazole, 5 % glycerol]. Eluted proteins were further purified and concentrated by ultrafiltration using Amicon Ultra 10K (for H-NS) and 30K (for DnaA) filters (Millipore). Finally, after overnight dialysis (0.025 μ m Membrane Filter; MF-Millipore) at 4 °C, the protein was stored in buffer C [45 mM Hepes-KOH, pH 7.6, 600 mM potassium glutamate, 1 mM DTT, 10 mM magnesium acetate, 0.5 mM EDTA, 20 % glycerol]. Judgement by SDS/PAGE with Coomassie Brilliant blue staining, the purity of each protein was >90 %. For the experiments where H-NS acts on the *datA* bound by DnaA, the purified DnaA, *datA* and ATP were first incubated in buffer D (20 mM Tris-HCl, pH 7.5, 150 mM potassium glutamate, 10 mM magnesium acetate, 8 mM DTT) at 20 °C for 10 min. Then the H-NS was added to buffer D (15 μ l) which contained DnaA-*datA* complexes and incubated for additional 10 min at 20 °C. The final complexes were separated by electrophoresis in 2 % (wt/vol) agarose gel at 80 V for 70 min in Tris-Borate buffer, followed by staining with SYBR Safe (Invitrogen).

Titration model

In the titration model, we posit that the cell mass grows exponentially according to the differential equation $\frac{dV}{dt} = \lambda V$, where V represents the volume of the cell and $\lambda = 1.54 \text{ hr}^{-1}$ (corresponding to doubling time of 27 min) is the growth rate. Concurrently, DnaA protein synthesis occurs at a rate proportional to the cell's mass, described by $\frac{dA}{dt} = \alpha_A V$, with $\alpha_A = 5 \text{ hr}^{-1} \text{ ml}^{-1}$ denoting the synthesis rate per unit volume. The chromosome's structure is abstracted as a vector representing the copy numbers of uniformly distributed sites along its length. Specifically, DnaA-boxes, which are binding sites for DnaA proteins, are quantified by summing the copy numbers at their respective locations. In this model, 300 DnaA-boxes (A_b) are evenly distributed on the chromosome while an additional 100 DnaA-boxes are located near the origin of replication. This arrangement approximates a head-weighted distribution of DnaA-boxes and helps prevent over-initiation (Hansen et al., 1991 [DOI](#)). Binding of DnaA proteins to these DnaA-boxes allows us to calculate the concentration of free DnaA as $[A_f] = (A - A_b)/V$, under condition of $(A - A_b) \geq 0$. DNA replication commences once the concentration of free DnaA exceeds a critical threshold $[A_f] = 10 \text{ ml}^{-1}$: $[A_f] > [A_f^*]$. 10 min sequestration time was set after replication initiation to mimic the sequestration time (Campbell & Kleckner, 1990 [DOI](#)).

To explore the dynamics of the *oriC* number following the cessation of *dnaA* expression, we can set the synthesis rate to be zero $\alpha_A = 0$. To simulate the dependence of initiation mass on *dnaA* expression, α_A varies from $1 \text{ hr}^{-1} \text{ ml}^{-1}$ to $64 \text{ hr}^{-1} \text{ ml}^{-1}$.

Extrusion model

The extrusion model was formulated following the titration model, with an additional extruder introduced as a protein with constant concentration $H = \alpha_H V$ with $\alpha_H = 100 \text{ ml}^{-1}$. The additional extruder competes the DnaA-boxes with DnaA proteins, with higher affinity to DnaA-boxes. So that it can displace DnaA from the DnaA-boxes, leading to a revised calculation for free DnaA concentration as $[A_f] = (A - (A_b - H))/V$, under condition of $(A_b - H) \geq 0$ and $(A - (A_b - H)) \geq 0$.

To get the DnaA repression simulation at the population level, we introduced cell division by the assumption of Cooper and Helmstetter that a cell divides at a constant time ($T_{\text{cyc}} = 60$ mins) after DNA replication initiation. Meanwhile, we introduced noises on cell mass growth rate and the DnaA synthesis rate at each cell division: $\lambda = \lambda_0 + \eta \lambda_0$, $\alpha_A = \alpha_0 + \eta \alpha_0$ where $\lambda_0 = 0.66 \text{ hr}^{-1}$, $\alpha_0 = 5 \text{ hr}^{-1} \text{ ml}^{-1}$ and $\eta = 0.1$ representing white noise. Symmetric cell division is used in the simulation for simplicity. This titration extrusion model is robust under various range of parameters.

DnaA-ATP/DnaA-ADP switch model

The switch model was simulated following the LDDR model in Berger *et al.* (Berger & Wolde, 2022 [DOI](#)). The same formulations of the DnaA-ATP fraction dynamics df/dt (equation 5) and the same parameters were used (Berger & Wolde, 2022 [DOI](#)). The cell division was determined when f reaches a threshold as defined previously (Berger & Wolde, 2022 [DOI](#)). To simulate the *dnaA* shut-down, we simply deleted the term of *dnaA* synthesis ($\lambda(1 - f)$).

Titration-switch model and titration-switch-extrusion model

In the integrated model combining titration and switch mechanisms, replication initiation is triggered when the concentration of free DnaA-ATP $[A_{ATP}^f]$ exceeds a threshold $[A_{ATP}^c]$. The concentration of free DnaA-ATP is titrated by the DnaA-boxes on the chromosome as in the titration model, with $[A_{ATP}^f] = (A_{ATP} - A_b)/V$, under condition of $(A_{ATP} - A_b) \geq 0$. The parameters related to the titration dynamics remain consistent with those described in earlier sections of titration model.

The dynamics of total DnaA-ATP (A_{ATP}) is governed by both expression and conversion processes as described in Berger *et al.* (Berger & Wolde, 2022 [DOI](#)). According to this framework, the lipid and DARS systems facilitate the conversion of DnaA-ADP to DnaA-ATP at constant rates, while the *datA* site and RIDA systems also play a role in converting DnaA-ATP to DnaA-ADP, again at fixed rates. Additionally, the synthesis of new DnaA proteins provides another source of DnaA-ATP. For the parameters governing DnaA-ATP dynamics, we adhere to those specified by Berger *et al.*, ensuring consistency with established research. To get a best fit of the initiation mass-*dnaA* expression relation, high threshold of free DnaA-ATP concentration was applied, $[A_{ATP}^c] = 200 \text{ ml}^{-1}$.

To simulate the titration-switch-extrusion model, we incorporated the extruder as described in the extrusion model, where its amount was given by $H = \alpha_H V$. The concentration of free DnaA-ATP was then calculated as $[A_{ATP}^f] = (A_{ATP} - (A_b - H))/V$, under condition of $(A_{box} - H) \geq 0$ and $(A_{ATP} - (A_b - H)) \geq 0$. To explore the best fit of continuous DNA replication rounds following *dnaA* shut-down, $\alpha_H = 400 \text{ ml}^{-1}$ is used.

Data availability

All data supporting the findings of this work are available from the corresponding author upon request. Source data are provided with this paper.

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Additional information

Code availability

Simulation data can be generated using the custom-made code and the parameter sets provided. The code is publicly available at https://github.com/BaiYangBqdq/dynamics_of_biomass_DNA_coordination .

Author contributions

C.L. and T.H. initiated and directed the research. D.L. and T.W. designed and evaluated the synthetic promoter library and performed mRNA FISH experiments with contributions from C.L. and H.Z. D.L., H.Z. and H.C. characterized the phenotype related to H-NS protein; D.L. and X.H. performed the DnaA shut down experiments; Y.B., H.Z., Z.Z. and T.H. conceptualized the extrusion model and performed the numerical simulations. All the authors analyzed the results and wrote the manuscript.

Additional files

Supplementary information 

Source data for figures 

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Reviewer #1 (Public review):

Summary:

The study by Li and coworkers addresses the important and fundamental question of replication initiation in *Escherichia coli*, which remains open, despite many classic and recent works. It leverages single-cell mRNA-FISH experiments in strains with titratable DnaA and novel DnaA activity reporters to monitor DNA activity peaks versus size. The authors find oscillations in DnaA activity and show that their peaks correlate well with the estimated population-average replication initiation volume across conditions and imposed dnaA transcription levels. The study also proposes a novel extrusion model where DNA-binding proteins regulate free DnaA availability in response to biomass-DNA imbalance. Experimental perturbations of H-NS support the model validity, addressing key gaps in current replication control frameworks.

Strengths:

I find the study interesting and well conducted, and I think its main strong points are:

- (1) the novel reporters obtained with systematic synthetic biology methods, and combined with a titratable dnaA strain.
- (2) the interesting perturbations (titration, production arrest, and H-NS).
- (3) the use of single-cell mRNA FISH to monitor transcripts directly.

The proposed extrusion model is also interesting, though not fully validated, and I think it will contribute positively to the future debate.

Weaknesses and Limitations:

- (1) A relevant limitation in novelty is that DnaA activity and concentration oscillations have been reported by the cited Iuliani and coworkers previously by dynamic microscopy, and to a smaller extent by the other cited study by Pountain and coworkers using mRNA FISH.
- (2) An important limitation is that the study is not dynamic. While monitoring mRNA is interesting and relevant, the current study is based on concentrations and not time variations (or nascent mRNA). Conversely, the study by Iuliani and coworkers, while having the drawback of monitoring proteins, can directly assess production rates. It would be interesting for future studies or revisions to monitor the strains and reporters dynamically, as well as using (as a control) the technique of this study on the chromosomal reporters used by Iuliani et al.
- (3) Regarding the mathematical models, a lot of details are missing regarding the definitions and the use of such models, which are only presented briefly in the Methods section. The reader is not given any tools to understand the predictions of different models, and no

analytical estimates are used. The falsification procedures are not clear. More transparency and depth in the analysis are needed, unless the models are just used as a heuristic tool for qualitative arguments (but this would weaken the claims). The Berger model, for example, has many parameters and many regimes and behaviors. When models are compared to data (e.g., in Figure 2G), it is not clear which parameters were used, how they were fixed, and whether and how the model prediction depends on parameters.

(4) Importantly, the main statement about tight correlations of peak volumes and average estimated initiation volume does not establish coincidence, and some of the claims by the authors are unclear in these respects (e.g., when they say "we resolve a 1:1 coupling between DnaA activity thresholds and replication initiation", the statement could be correct but is ambiguous). Crucially, the data rely on average initiation volumes (on which there seems to be an eternally open debate, also involving the authors), and the estimate procedure relies on assumptions that could lead to biases and uncertainties added to the population variability (in any case, error bars are not provided).

(5) The delays observed by the authors (in both directions) between the peaks of DnaA-activity conditional averages with respect to volume and the average estimated initiation volumes are not incompatible with those observed dynamically by Iuliani and coworkers. The direct experiment to prove the authors' point would be to use a direct proxy of replication initiation, such as SeqA or DnaN, and monitor initiations and quantify DnaA activity peaks jointly, with dynamic measurements.

(6) While not being an expert, I had some doubt that the fact that the reporters are on plasmid (despite a normalization control that seems very sensible) might affect the measurements. Also, I did not understand how the authors validated the assumptions that the reporters are sensitive to DnaA-ATP specifically. It seems this assumption is validated by previous studies only.

Overall Appraisal:

In summary, this appears as a very interesting study, providing valuable data and a novel hypothesis, the extrusion model, open to future explorations. However, given several limitations, some of the claims appear overstated. Finally, the text contains some self-evaluations, such as "our findings redefine the paradigm for replication control", etc., that appear exaggerated.

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Reviewer #2 (Public review):

Summary:

The authors show that in *E. coli*, the initiator protein DnaA oscillates post-translationally: its activity rises and peaks exactly when DNA replication begins, even if *dnaA* transcription is held constant. To explain this, they propose an "extrusion" mechanism in which nucleoid-associated proteins such as H-NS, whose amount grows with cell volume, dislodge DnaA from chromosomal binding sites; modelling and H-NS perturbations reproduce the observed drop in initiation mass and extra initiations seen after *dnaA* shut-down. Together, the data and model link biomass growth to replication timing through chromosome-driven, post-translational control of DnaA, filling gaps left by classic titration and ATP/ADP-switch models.

Strengths:

(1) Introduces an "extrusion" model that adds a new post-translational layer to replication control and explains data unexplained by classic titration or ATP/ADP-switch frameworks.

(2) A major asset of the study is that it bridges the longstanding gap between DnaA oscillations and DNA-replication initiation, providing direct single-cell evidence that pulses of DnaA activity peak exactly at the moment of initiation across multiple growth conditions and genetic perturbations.

(3) A tunable dnaA strain and targeted H-NS manipulations shift initiation mass exactly as the model predicts, giving model-driven validation across growth conditions.

(4) A purpose-built Psyn66 reporter combined with mRNA-FISH captures DnaA-activity pulses with cell-cycle resolution, providing direct, compelling data.

Weaknesses:

(1) What happens to the (C+D) period and initiation time as the dnaA mRNA level changes? This is not discussed in the text or figure and should be addressed.

(2) It is unclear what is meant by "relative dnaA mRNA level." Relative to what? Wild-type expression? Maximum expression? This should be explicitly defined.

(3) It would be helpful to provide some intuition for why an increase in dnaA mRNA level leads to a decrease in initiation mass per ori and an increase in oriC copy number.

(4) The titration and switch models do not explicitly include dnaA mRNA in the dynamics of DnaA protein. Yet, in Figure 2G, initiation mass is shown to decrease linearly with dnaA mRNA level in these models. How was dnaA mRNA level represented or approximated in these simulations?

(5) Is Schaechter's law (i.e., exponential scaling of average cell size with growth rate) still valid under the different dnaA mRNA expression conditions tested?

(6) The manuscript should explain more explicitly how the extrusion model implements post-translational control of DnaA and, in particular, how this yields the nonlinear drop in relative initiation mass versus dnaA mRNA seen in Figure 6E. Please provide the governing equation that links total DnaA, the volume-dependent "extruder" pool, and the threshold of free DnaA at initiation, and show - briefly but quantitatively - how this equation produces the observed concave curve.

(7) Does this Extrusion model give well well-known adder per origin, i.e., initiation to initiation is an adder.

(8) DnaA protein or activity is never measured; mRNA is treated as a linear proxy. Yet the authors' own narrative stresses post-translational (not transcriptional) control of DnaA. Without parallel immunoblots or activity readouts, it is impossible to know whether a six-fold mRNA increase truly yields a proportional rise in active DnaA.

(9) Figure 2 infers both initiation mass and oriC copy number from bulk measurements (OD_{600} per cell and rifampicin-cephalexin run-out) instead of measuring them directly in single cells. Any DnaA-dependent changes in cell size, shape, or antibiotic permeability could skew these bulk proxies, so the plotted relationships may not accurately reflect true initiation events.

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